

[DC] Animating Interactive Virtual Humans

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ABSTRACT

Immersive virtual worlds, such as in video games, have captivated the general public for years. Virtual reality (VR) takes immersion further, placing users inside virtual environments and allowing them to interact with virtual objects and even socialize with others in a shared virtual space. The effectiveness of VR is heavily influenced by the quality of virtual interactions and characters. This research proposal aims to address some of the challenges of creating engaging virtual experiences by focusing on the motion of virtual humans.

Index Terms: Computer Graphics—Virtual Reality—; Computer Graphics—Animation

1 INTRODUCTION

A significant challenge when creating interactive virtual humans is controlling how they move in response to user input, whether in video games or VR. For offline uses, such as film and video game cut-scenes, detailed marker-based motion capture (mocap) is widely used. However, due to the dynamic nature of real-time applications, mocap imposes significant constraints because the motions are often pre-recorded and not easily adaptable to different scenarios. In immersive VR, users often control an embodied avatar to interact with a virtual environment. Most modern consumer VR devices include head and hand tracking, which facilitate basic interactions. However, there are limitations to the current state of motion tracking in VR. For example, there is no tactile feedback when interacting with virtual objects without controllers. When it comes to interacting and conversing with others in VR, gaze animation remains rudimentary and repetitive when eye tracking is not an option. In this proposal, we address several issues related to the animation of virtual humans and avatars for real time applications, such as ensuring that movements respond appropriately when avatars interact with virtual objects and eye motions are appropriate when engaging with other avatars.

2 FEEDBACK FOR VIRTUAL GRASPING

Interacting with virtual objects is a significant part of the VR experience, and thus has been the focus of a great deal of research. There have been many proposed methods for interacting with virtual objects, but using one's hands may be the most intuitive and natural [5, 1]. Despite the potential advantages of hand tracking based interaction, there are several challenges and limitations that arise in practice. One limitation when interacting with virtual objects via hand tracking in VR is that there is no tactile feedback. Furthermore, the virtual hands may intersect with solid virtual objects, potentially leading to a degraded experience. To address these limitations, we examined the effectiveness of different visual feedback techniques for virtual grasping [4]. Some of our conditions involved augmenting the motion of the virtual hand such that it remained outside of the virtual object ("outer hand", Figure 1), even if the user's tracked hand entered it. In a within subjects experiment, we measured grasp and release performance as well as effects on ownership. We found that

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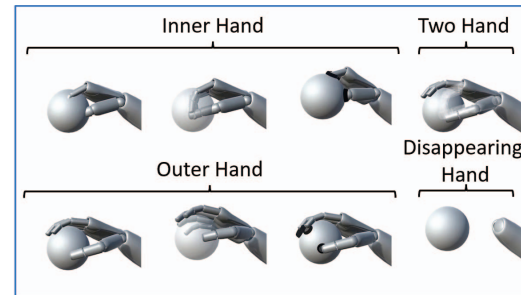


Figure 1: The eight visual feedback techniques for grasping implemented in our study. The top row consists of the three "inner hand" feedback visualizations, where the virtual hand always follows the tracked hand, even into the object. The bottom row shows the three "outer hand" feedback visualizations, where the virtual hand could not enter the virtual object, even if the tracked hand does. "Two hand" combines "inner hand" and "outer hand", and "disappearing hand" hides the hand while grasping.

users preferred "outer hand" feedback visualizations, even though they resulted in lower grasping performance. We also found some evidence that "outer hand" motions could result in higher levels of ownership. We additionally conducted a similar study in which we included audio feedback as a condition [3]. We limited our grasping visualization to "outer hand", and manipulated the combinations of visual and audio feedback within-subjects. Interestingly, we found that audio feedback reduced performance but was the most preferred by users. We also found indications that having both visual and audio feedback might increase ownership.

3 AVATAR GAZE SYNTHESIS FOR COMMUNICATION IN VR



Figure 2: The avatars used in our user study evaluating our method for synthesizing conversational gaze.

Gaze is an integral part of communication, serving as an essential non-verbal cue that conveys interest, personality, and more [6]. With an increasing user base for avatar-mediated communication (e.g. VR Chat), there is a growing demand for expressive avatars. Therefore, we developed a method to augment avatar expression by synthesizing the eye motions during two-party conversations [2]. We collected

head and eye motion data along with audio to develop a sequence model that can infer the gaze direction of an avatar based on the audio and head movement data of the user. We evaluated our model by conducting a study in which participants viewed short clips of avatars involved in conversation (Figure 2), and rated qualities of the eye animation such as naturalness and engagement. We also included eye animation from a procedural model, the recorded eye animation, and no eye animation in the study for comparison. Results suggest that simple gaze models that generally maintain an acceptable amount of eye contact, such as the procedural model, may suffice for short conversations. The study also provided some insight into how people subjectively rate eye motion for conversation, with results suggesting that people are sensitive to errors in eye animation and prefer when there is more eye contact and less eye movement.

4 PROPOSED WORK FOR HAND MOTION SYNTHESIS



Figure 3: We are in the process of collecting high quality motion data of guitar playing.

Animating the hands of virtual humans continues to be a significant challenge. Motion capture of the hands using traditional marker-based motion capture technology is expensive and time-consuming. Hand tracking methods commonly used in VR devices are not very precise, and due to the placement of the hand tracking cameras, occlusion is often an issue.

Motivated by the significant challenges of animating hands, we propose a hand motion synthesis algorithm for guitar playing. The complexities addressed will be the ambiguity of which finger to use to play certain notes or chords, the high variability of motion, and achieving natural looking motion.

Our proposed method involves motion capture of hand motions during guitar performance (Figure 3). Our captured motion data will be used to develop a model that can realistically simulate hand movement while playing guitar. The model will take written guitar music as input and automatically animate virtual hands to “play” the input song. This project will enable detailed hand animation based on written music without the need for motion capture and could be extended to other instruments. Applications include virtual music performance, such as VR concerts, the creation of music videos or movies with virtual characters playing instruments, and could even be extended to other applications involving complex hand movement outside of music.

5 CONCLUSION

My research has broadly covered virtual humans and animation. I welcome constructive feedback and insights regarding the proposed hand animation project. Specifically, I seek suggestions on extending the proposed project into applications or user studies within virtual reality (VR) or mixed reality (MR) contexts.

Additionally, guidance on promising directions to take my research in the future would be greatly appreciated. My core interests lie at the intersection of VR technology, motion synthesis, and motion perception.

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